HUBERT KIM

AUTOMATION RESEARCH ENGINEER

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CAPABILITIES

- Systematic integration of multi-robot planning and sequencing
- Setting up Boundary Conditions for motion planning involving physical contact
- Quantitative analysis of 2D /3D image for GR&R

ACHIEVEMENTS

GE Aerospace Research |

Spotlight Award	Apr '25
Patent Award	Jan '25
Applause Award	Aug '24
Bravo Award	Oct '23

ICTAS, Virginia Tech |
Doctoral Scholarship 2016–20

EDUCATION

PhD,
Mechanical Engineering
Virginia Tech, *Blacksburg, VA*
Earned in Dec 2021

BS, *cum laude,*
Mechanical Engineering
NYU Tandon, *Brooklyn, NY*
Earned in May 2015

SUMMARY

Data-driven robotics engineer reducing Direct Labor hours via automation

- Defining problems and questions of how to transform an expert's motion to robot motion.
- Three years of experience in teams with early failure & learn (TRL 1 → 5).

Mechatronics | Robot Perception & Path Planning | Signal Processing

PROFESSIONAL EXPERIENCE

AUTOMATION/MECHATRONICS ENGINEER

Mechanical Components and Systems Lab | **GE Aerospace Research Center**

Automated Layup Composite project — Aug '22 – Current

: Product cost savings and part assembly quality improvements through automation of the manufacturing system.

- Established a multi-robot cell with a Python HMI platform, providing coordinated robot motion with an offline trajectory planning feature.
- Identified and analyzed risks during pick and place tasks to reduce the probability and impact of the root causes.
- Improved the precision by controlling the error boundary during the pick and place task on the 3D contour surface.

Surveillance Robot Inspect project — Aug '22 – Aug '23

: Prototyped a surveillance robot to solve the accessibility problem under a confined space while keeping its structural rigidity (IP filed & Patent Award)

- Practiced quick & dirty approaches (Most Viable Products) for characterizing robot structure & materials, minimizing deflection during deployment.
- Developed a pixel segmentation method for evaluating the robot deflection.

GRADUATE RESEARCH ASSISTANT — May 2015–21

Assistive Robotics Laboratory | **Virginia Tech** — Advisor: Dr. Alan T. Asbeck

Studied a wearer's perceptual response during motion training via exoskeleton. The quantitative studies led to the discovery of the human perceptual threshold as 0.1—0.2 Nm for the arm under external loading, and 0.4—0.8 Nm for the arm under motion, as represented in *Scientific Reports*

Dissertation: "Joint Torque Feedback for Arm Motion Training"

UNDERGRADUATE RESEARCH ASSISTANT — May 2013–15

Dynamic System Laboratory | **NYU** — Advisor: Dr. Maurizio Porfiri

Applied frequency domain system identification & impedance analysis to the Polymer Composite to characterize its properties and validate a physical model.

Thesis: "Voltage attenuation along the electrodes of ionic polymer metal composites"